

DGN4500 - Design Thinking Approaches

Naomi (Xiao Li) 35963328

BLOOMLY

A Symbiotic Garden App
for Emotional Interaction

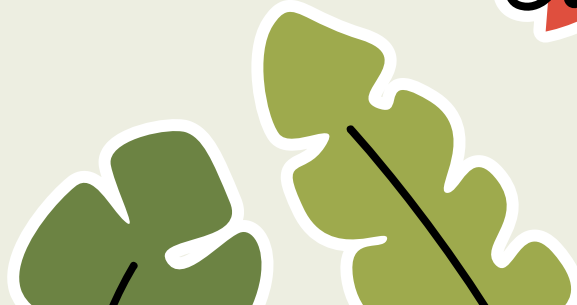
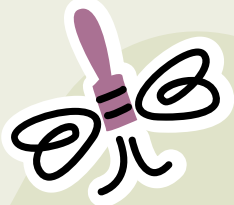


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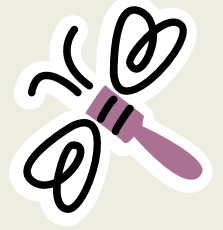
Preliminary Idea

03

**Process
development**

04

**Final work
and feedback**





I like Chinese food the most, especially spicy food.

I know it's not good to play mahjong all night, but it's really interesting.

I always have endless topics to talk about with my friends.

It's boring to call my parents, but I have to do it.

SAY

DO

Play mahjong all night every week

Clean the room every week

The routine is very irregular (Especially eating habits)

I will go out to eat and play with my good friends.

Call parents every week

Exercise occasionally every week

Empathy Map

● Positives ● Negatives

I need to make my life more regular.

Even I am happy to get along with friends, sometimes I still feel lonely.

I want to get more parents' responses.

I'm trying to complete my study in a more planned way.

I want to go on a trip, but just think about it. Many things are like this.



THINK

FEEL

Going out with friends or playing mahjong makes me very happy.

I feel a little cramped when I call my parents, because they will only keep asking me questions.

The pressure of schoolwork makes me tired.

Lazy

Stressful

Although my parents' calls are boring, their calls make me feel at ease.



Story :

Jasmine is a 23-year-old international student from Sichuan, China. She's vibrant, confident, and beautiful—someone who brings a spark of energy to every room. Since moving to Australia for her studies, she's built strong friendships and often spends her nights laughing and playing mahjong with her friends until sunrise.

But in the quiet moments, Jasmine sometimes feels a little lonely. She misses home, especially her parents who love her deeply and support her education abroad. They often ask her to call, and while she always does, these conversations tend to feel one-sided. Jasmine shares stories, but her parents often just listen silently or repeat the same few questions. She knows they care, but the lack of emotional feedback leaves her feeling unheard.

She loves spicy Sichuan food and rarely wakes up before noon. Her lifestyle is carefree and a little chaotic, but she's trying to build meaningful connections—even from a distance.

Something unique :

Jasmine values emotional connection and believes relationships should be mutual. She expects care and effort to be returned, and feels most engaged when there's warmth, response, and emotional presence.

Character Map



Jasmine

Tag

Emotionally Invested

Loves Doing Things Together

Social & Fun-Loving

Hates One-Way Communication

Flexible Lifestyle

Something she think is great :

- Having flexibility and freedom in how she does things
- Getting responses—she values feedback and being heard
- Doing things together with close friends
- Discovering fresh, new, and fun experiences

Something she hate :

- Putting in effort but getting no result or response
- Being forced into strict rules or fixed routines
- One-way conversations or emotional coldness
- Feeling controlled or emotionally drained

Powers :

Jasmine is easily drawn to platforms like Red Note because the content feels endlessly fresh and relatable—it makes her feel inspired.

She loves virtual pet games and apps that involve growing, because they give her a sense of achievement and ongoing responsibility.

Need :

A more regular life

A more balanced way to feel emotionally connected with her parents—something that helps express affection and receive emotional feedback without relying on forced conversations.

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Design direction

Design a gentle, shared experience that helps Jasmine and her parents feel emotionally connected without relying on forced conversations.

Pain point

Emotional one-way communication

She often feels like she's the only one sharing during calls with her parents, with little response or emotional feedback.

Dislikes forced interactions

Jasmine resents rigid or guilt-driven obligations to call, especially when they feel like chores rather than real connections.

HMW

How Might We Questions

HMW help Jasmine connect with her parents without small talk?

HMW make long-distance communication feel more mutual?

HMW make everyday interactions feel meaningful?

HMW create a shared experience across distance?

HMW encourage bonding in a flexible, gentle way?

Amp up the good

HMW make the joy of response (like getting a message) more rewarding?

ID unexpected resources

HMW use ambient objects (like the light-up desk plant) to deliver silent emotional feedback?

Remove the bad

HMW avoid awkward small talk without cutting off communication?

Create an analogy from need or context

HMW design family interaction like co-op gameplay with visible progress?

Explore the opposite

HMW make fewer interactions feel more powerful?

Play POV against the challenge

HMW make Jasmine want to interact because it feels fun and light, not like a chore?

Question an assumption

HMW create intimacy without verbal expression?

Break POV into pieces

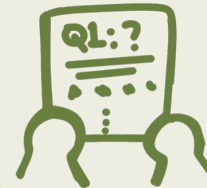
HMW help parents feel seen and involved without daily calls?

HMW help Jasmine connect with her parents without small talk?

HMW make the joy of response (like getting a message) more rewarding?



Preliminary Idea





1. Send photos instead of just calling



2. Have a common topic



3. Touch through something, - like iWatch 'Digital Touch'



4. Respond through light



5. Express love directly



6. Raise a digital pet together



7. Record every call



8. Plant a digital garden with your family



9. Shared digital calendar



10. Make a memory puzzle with your family



11. Let them know your life and the culture of your location.



12. Make the reply interesting



13. Make a time capsule and keep love.



14. Do the question table with your family



15. Respond to messages with stickers (Simple way)



Main Ideas



8. Plant a digital garden with your family

I want to create a **relationship garden**, where you and your family can co-grow a plant — calls become water, photo sharing becomes sunlight, and mutual replies become the fertilizer that helps it thrive.

Functions
Ideas
include



15. Respond to messages with stickers (Simple way)



12. Make the reply interesting



7. Record every call



1. Send photos instead of just calling



2. Have a common topic

Character Map 2



Jasmine's mother

Ms. Yang

Age : 49

Planting enthusiasts

Short video lovers

Story :

Yang is a 49-year-old plant enthusiast from Sichuan who grows a new plant every season with care and joy. She loves watching short videos with life advice and often shares them with her daughter. Sometimes she plays WeChat mini games for fun. Though not always talkative, she asks her daughter to call at least three times a week — for her, it's how love stays close, even from far away.

Something unique :

She expresses love through care and routine — whether it's tending to plants or gently reminding her daughter to call home.

Need :

· A gentle, meaningful way to feel emotionally connected with her daughter
· Feedback or signs of appreciation from loved ones
· A simple tool that fits her daily habits and doesn't feel "too technical"

Something she think is great :

Watching short videos with life lessons
Growing seasonal plants
Getting regular phone calls from her daughter
Light games that are simple but fun

Prototype -1



UX/UI Design

I made a prototype of a program and sorted out the logic of the whole app.



Material Attempt



Illumination

Water the plant

Plant



Test – 1 & Improvement of ideas

Vibe test



Humidifier object

Following insights from Test 1, I decided to shift away from the original idea of offering a discount on real plant purchases once two co-growers reached a high intimacy level. Instead, I'm moving toward a more immersive and emotionally engaging approach — generating unique **virtual plants** and expressing relationship growth through a **physical model plant**.



Beyond the app, I now want to incorporate **physical feedback** into the model plant itself, so that interactions can be felt even without checking the phone. This aligns with the goal of creating an emotional connection that feels ever-present in daily life.

Prototype - 2

Bloomly Grow

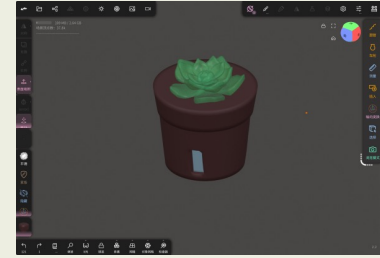


Model flower =
Transparent material
↳ can glow

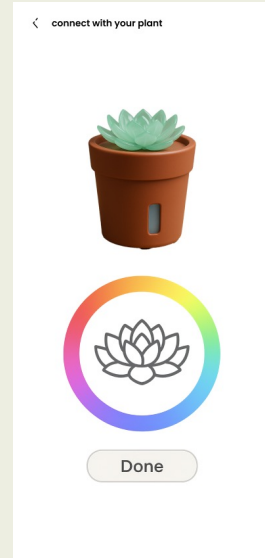
Humidifier.



I used
Nomad
to model.



I added a light control UI
to the initial prototype,
allowing users to
connect with the
physical model plant.
Different light colors
represent different
meanings, enabling
users to receive
emotional feedback
through the plant's
glowing response.

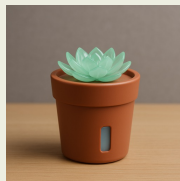
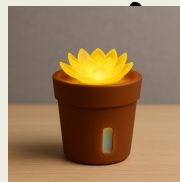
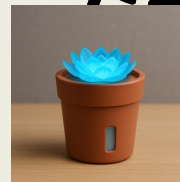
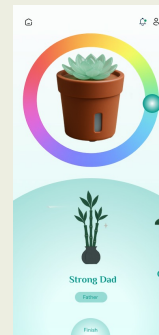
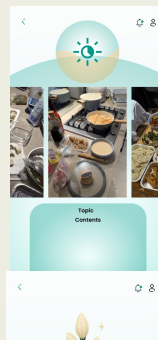
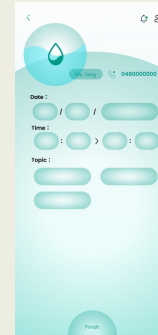
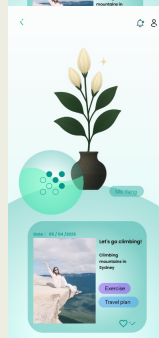
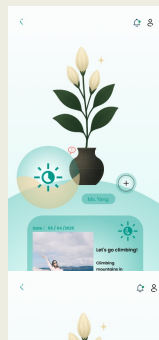
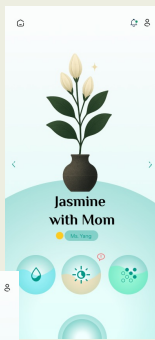
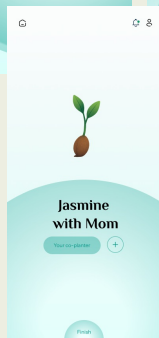
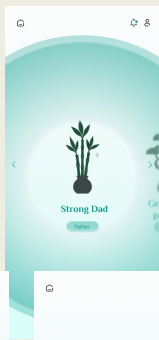
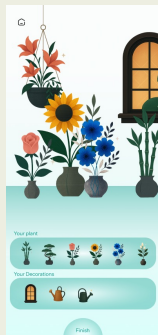


User research

Based on user research, I found that Jasmine has a habit of using a humidifier. Her current device holds 1000ml of water and only needs to be refilled about once every two weeks. To subtly encourage her to think about the app and the plant she is co-growing with her family, I decided to reduce the tank capacity to 300ml — requiring a refill roughly once a week. This smaller size not only maintains a compact, desk-friendly form but also serves as a gentle physical reminder to reconnect.



Final App



Test link:

<https://www.figma.com/proto/FFj6qybIJNQ6apulKMnXyG/Untitled?content-scaling=fixed&kind=proto&node-id=1-3&page-id=0%3A1&scaling=scale-down&starting-point-node-id=1%3A3>

Bloomly Grow's color

• When Bloomly Grow lights up, what does it mean?

Each **light color** on your Bloomly Grow represents a **different person** you're co-growing a plant with.

When the plant flashes a specific color, it means that person just interacted with you — maybe they shared a photo, replied to your message, or made a call.

The **glowing light** is their way of saying:

“I'm here. I'm thinking of you.”

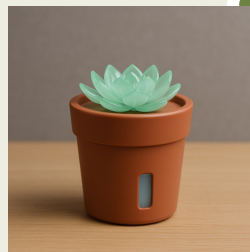
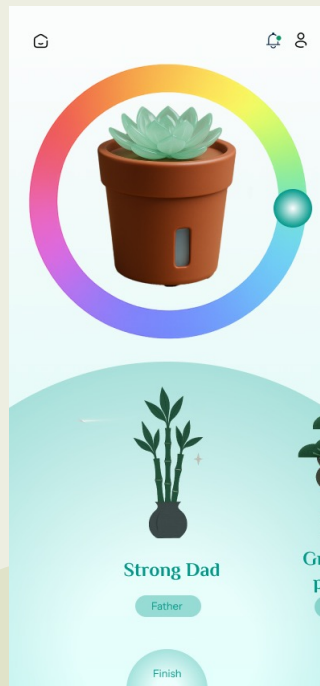
Because Bloomly lets you grow multiple plants with different people — your family, friends, partner, or anyone close to your heart — each plant has its **own color**, making every bond feel unique and visible in your daily space.

• Why is there a lot of color in Bloomly Grow? And why can you choose a colour for the plant by yourself?

In Bloomly Grow, every plant represents someone important to you. By letting you choose the color yourself, we give you a way to make each connection **truly personal**.

Whether it's soft green for your mom or bright pink for your best friend — you define what each color means.

That way, when the light appears, you'll know **exactly who's thinking of you**, without needing a single word.





Test2 & feedback

Test method

1. Teach the prototype tutorial to Jasmin and her mother to make sure that they understand how to use it.



2. Use WeChat to simulate recording and use jasmine's humidifier. Put the humidifier in front of the table and test the frequency of use.



Questionnaire (For Jasmine and her mother)

Version A: Jasmine – The Daughter (Age 23, international student)

Section 1: Overall Experience

1. Did you enjoy the idea of growing a virtual plant with your parent less emotionally meaningful to you?
- ☐ Yes, very much
 - ☐ A little
 - ☐ Not really
 - ☐ No

Section 2: The Physical Model

4. How did you feel when the mode → Short answer
5. Did the physical plant help you feel emotionally connected to your parent when you weren't checking your phone?
- ☒ Yes, definitely
 - ☐ Maybe
 - ☐ Not really
 - ☐ No

Section 3: Interaction Patterns

7. How easy or natural did it feel to use calls, photos, and replies to care for the plant?
- ☒ Very easy
 - ☐ A little difficult
 - ☐ I needed help
 - ☐ I didn't like to use this plant

Section 4: Final Thoughts

10. What part of the experience did you enjoy most?
11. What would you improve to make this experience more natural or emotionally strong?
- Paragraph

Version B: Yang – Jasmine's Mother (Age 49, plant lover)

Section 1: Overall Experience

1. Did you enjoy the idea of growing a plant together with your daughter through daily interactions?
- ☒ Yes, it felt special
 - ☐ It was okay
 - ☐ Not really
 - ☐ I didn't fully understand it

Section 2: The Model Plant

4. How did you feel when the model plant lit up or responded to interaction?
- Short answer
5. Did the plant's light or change make you feel your daughter was thinking of you?
- ☒ Yes, very much
 - ☐ A little
 - ☐ Not really
 - ☐ I didn't notice

Section 3: Interaction and Effort

7. How easy or natural did it feel to use calls, photos, and replies to care for the plant?
- ☒ Very easy
 - ☐ A little difficult
 - ☐ I needed help
 - ☐ I didn't like to use this plant

Section 4: Final Thoughts

10. What did you enjoy the most about this experience?
- Paragraph
11. What part would you like to make easier or clearer?
- Paragraph

版本 B: 妈妈 (Yang, 49岁, 植物爱好者)

第1部分: 整体体验

1. 您是否喜欢通过日常互动和女儿一起种植一株植物的想法?
- ☒ 非常喜欢, 感觉很棒
 - ☐ 还可以
 - ☐ 不太喜欢
 - ☐ 完全没理解这个概念

第2部分: 实体模型植物

4. 当您看到模型植物亮起灯或发出声音时, 您有什么感觉?
- ☒ 很开心, 感觉女儿在想我
 - ☐ 没什么感觉
 - ☐ 有点好奇
 - ☐ 不太喜欢

第3部分: 使用与互动

7. 您觉得用这个App发照片、打电话、回复消息来照顾植物容易吗?
- ☒ 非常容易
 - ☐ 有点困难
 - ☐ 需要帮助
 - ☐ 我不喜欢用这个植物

第4部分: 开放反馈

10. 在这个体验中, 您最喜欢的部分是什么?
- 简短回答
11. 您觉得哪些方面希望能更简单或更清晰?
- 简短回答

Storyboard & Summary



In study and work



The device lights up



You can choose:

Look at who sent you miss/
Continue to work, but received
longing from afar.



Take the initiative
to share in the app



I have more
common topics
with my family.



Two-way sharing
makes the emotional
connection closer.

Feasibility

- Users responded positively to the theme of growing something together.
- The humidifier function makes it easy to integrate into daily life.
- Replaces small talk with meaningful actions like calls and photo sharing.

Issues

- The app interaction needs improvement, especially for older users.
- AR content is not yet developed, making the reward less engaging.
- Light feedback from the model plant may feel abstract and needs clearer emotional meaning.

Thanks!

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